Evaluating Ultra-Low-Energy Direct Air Capture, Carbon Storage, and Waste-Heat Integration Pathways for Managing Dynamic Cooling and Energy Needs of Large Electrical Loads

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Period of Performance: 5 Months; Total Budget: $70,000

Background

Utilities across the Southwest are confronting unprecedented electric-load growth driven by extreme heat, electrification, and the rapid expansion of hyperscale and AI-optimized data centers. These facilities operate continuously, draw hundreds of megawatts per campus, and place increasing demands on electricity generation and transmission infrastructure. Forecasts suggest AI data-center electricity demand could grow dramatically over the coming decade, creating significant challenges for utilities seeking to maintain reliability while supporting continued economic growth. At the same time, data-center operators are pursuing net-zero goals while seeking 24/7 renewable matching. Carbon dioxide removal (CDR) technologies such as direct air capture (DAC) can help offset residual emissions, but their cost and interactions with energy systems depend strongly on the requirements for air movement, sorbent regeneration, and product gas conditioning. Process configurations that decouple atmospheric capture from regeneration and compression may allow the more energy-intensive operations to be scheduled around low-cost electricity and suitable heat sources. Data centers may offer opportunities for such integration because they reject substantial amounts of low-grade heat, although its usefulness depends on temperature level, temporal availability, and site-specific process requirements. This project will therefore assess whether combining low-energy capture, intermediate carbon storage, and waste-heat-assisted CO_2 recovery can provide meaningful carbon-management and load-flexibility benefits, while evaluating cooling and energy-storage services as potential co-benefits.

A key innovation developed by ASU CNCE is an ultra-low-energy DAC architecture that uses passive air-contacting systems combined with intermediate liquid (bi)carbonate storage to accumulate captured atmospheric CO_2 with minimal electricity consumption. Unlike conventional DAC systems that require regular energy-intensive sorbent regeneration, compression, and purification steps, this approach stores captured carbon in low-cost liquid media and decouples atmospheric capture from energy-intensive upgrading operations. Previous CNCE work further demonstrated pathways for progressively increasing CO_2 partial pressure through a ladder of intermediate carbonate and bicarbonate storage media using low-grade heat, creating the potential to recover and upgrade stored carbon using waste heat and compress it when low-cost off-peak or renewable electricity is available (**Figure 1**).

The compression and conditioning infrastructure required for product gas transport and sequestration may provide a starting point for evaluating additional liquid-CO_2 storage and

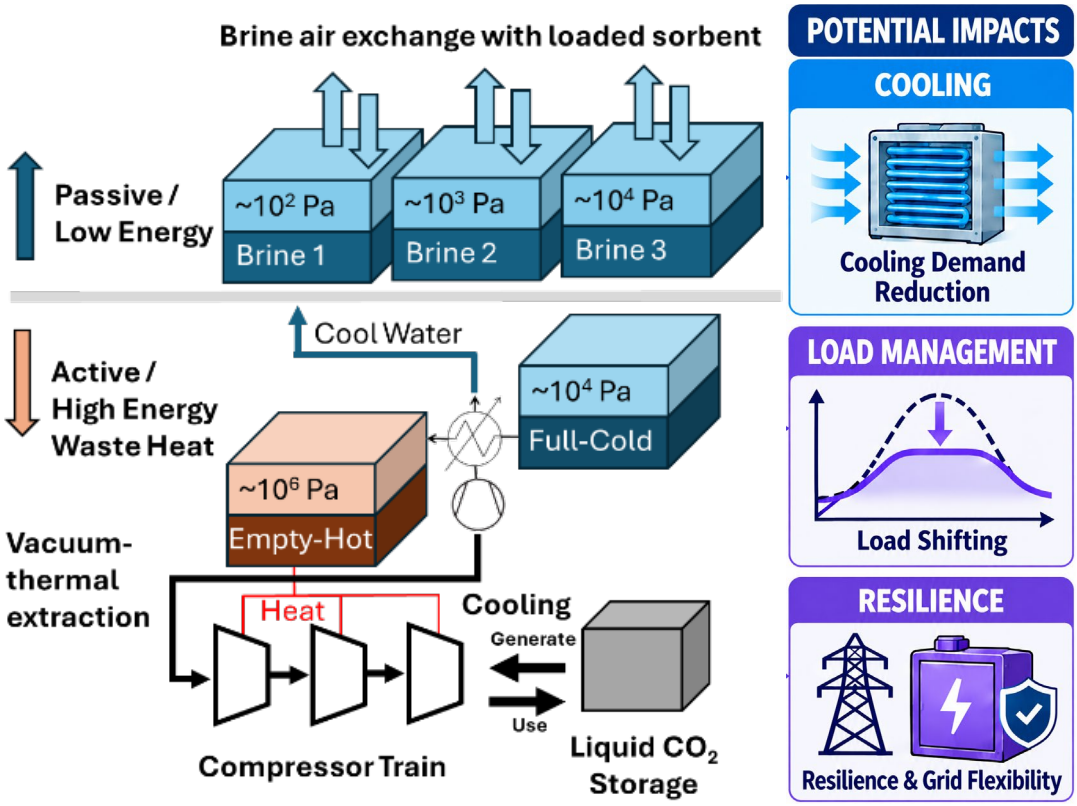


Figure 1. Illustration of how humid air from the brine storage is used to release CO_2 from the loaded sorbent and captured into intermediate storage in (bi)carbonate brine tanks and subsequently extracted and compressed to liquid CO_2 when intermittent waste heat or low-cost power is available. Potential additional benefits are load shifting and reduced cooling-related electricity demand.

cooling services. Liquid CO_2 produced during periods of surplus or low-cost electricity could serve as an intermediate energy-storage medium. In addition, controlled expansion could produce a cold CO_2 stream that, through an indirect heat exchanger coupled to the data center's existing cooling-water loop, may reduce cooling-related electricity demand during periods of elevated ambient temperatures. The assessment will determine whether any resulting load-shifting benefit justifies the additional liquefaction, storage, expansion, and recompression requirements.

This project will evaluate the technical feasibility, operating requirements, and economic viability of selected integration pathways among ultra-low-energy CO_2 capture, intermediate carbon storage, waste-heat-assisted CO_2 upgrading, liquid CO_2 storage, and large-load energy-management, including their potential cooling and electric load-management.

Objective 1 – Evaluate Ultra-Low-Energy Wind-Driven Atmospheric CO_2 Capture

Objective 1 will evaluate an ultra-low-energy atmospheric CO_2 capture system that uses natural wind to drive airflow through a turbine and a rectangular sorbent-contacting vessel containing stacked sheets of mesh-encapsulated ion-exchange sorbent beads (**Figure 2**).

Computational fluid dynamics (CFD) modeling and experimental validation will quantify the effects of sorbent-sheet spacing, packing density, contactor length, pressure drop and wind speed on airflow distribution, . Combined with independently specified sorbent equilibrium and kinetic parameters, the resulting hydrodynamic model will support screening estimates of cyclic CO_2 working capacity and capture performance.

Task 1.1 – CFD Modeling of Wind-Driven Sorbent Contactors

A CFD model will be developed in Ansys Fluent using a porous-medium representation of the sorbent sheets within a rectangular contactor geometry. The model will simulate airflow through stacked layers of mesh-contained sorbent beads and evaluate:

- Pressure drop as a function of airflow rate
- Effects of sorbent-sheet spacing and packing density
- Effects of contactor length and flow-path geometry
- Airflow distribution and bypassing
- CO_2 and water-vapor transport through the contactor, using specified sorbent parameters
- Screening estimates of contactor productivity under representative operating conditions

The CFD model will identify hydrodynamically feasible configurations and quantify tradeoffs among airflow distribution, contactor dimensions, sorbent packing, and pressure drop.



Figure 2. Images of a (top) a mesh packet-containing sorbent beads (bottom) and several mesh packets loaded in a low pressure drop contactor for evaluation in the wind tunnel.

Task 1.2 – Experimental Validation

Existing wind-tunnel infrastructure and contactor prototypes (Figure 2 bottom) will be used to experimentally validate CFD predictions. Pressure drop and airflow rate will be measured across multiple sorbent-sheet spacings and contactor depths using mesh-contained sorbent beads. Experimental results will be used to calibrate porous-medium parameters and validate airflow predictions across the anticipated operating range. A small prototype will be built using a commercial whirlybird roof vent connected to a series of sorbent contactor modules coupled together to validate the pressure drop and flows through extended contactor form factors and identify optimal contactor module sizes for pilot-scale systems.

Task 1.3 – Contactor Performance Assessment

The validated hydrodynamic model, combined with specified sorbent equilibrium and kinetic data, will be used to estimate contactor performance. The assessment will estimate:

- Sorbent working capacity under specified adsorption and regeneration conditions
- CO_2 capture rates as a function of airflow rate (wind speed)
- Impact of temperature and relative humidity on sorbent performance
- Contactor productivity (kg CO_2 captured per contactor volume and time)
- Tradeoffs between capture efficiency and airflow resistance

The resulting model will provide performance inputs for subsequent process-feasibility, energy-storage, cooling, and load-management analyses.

Deliverables

D1.1: Validated hydrodynamic CFD model of airflow distribution and pressure drop in a wind-driven (passive) contactor based on a rectangular sorbent-containment vessel and porous-medium representation of mesh-contained sorbent sheets (Month 3).

D1.2: Experimental validation of pressure-drop and airflow across multiple sorbent-sheet spacings and contactor geometries (Month 4).

D1.3: Screening model relating ambient wind conditions, turbine and contactor characteristics (pressure drop, flow patterns), ambient temperature and humidity, contactor geometry, and expected CO_2 capture productivity (Month 5).

Milestone

M1.1: Identify a wind-driven contactor configuration capable of maintaining ultra-low pressure drop (≤ 10 Pa) while achieving sufficient airflow and predicted CO_2 capture productivity under representative Southwestern operating conditions (≤ 2 m/s average wind) (Month 5).

Objective 2 – Process Evaluation of Carbon Storage, Cooling, and Load-Management Pathways

Building on Objective 1, Objective 2 will assess whether captured CO_2 can be transferred into intermediate liquid storage, progressively concentrated through waste-heat-assisted absorption and regeneration steps, and conditioned for compression or liquefaction under technically and economically credible operating conditions. Rather than assuming a fully integrated architecture, the objective will identify practical operating envelopes, thermodynamic constraints, equipment requirements, and the most promising pathways for future development, while focusing on meaningful carbon-management and load-management value, with cooling evaluated as a potential additional benefits. Initial analyses will assume idealized equilibrium transfer and sufficient heat and mass transfer between

process steps to establish theoretical performance bounds and corresponding process-performance requirements.

Task 2.1 – Process Modeling and Waste-Heat Integration

Develop process flow diagrams and mass- and energy-balance models describing transfer of CO_2 from the regenerated solid sorbent into intermediate carbonate storage, staged waste-heat-assisted CO_2 concentration and recovery, product gas conditioning, and subsequent compression and liquid CO_2 production. The process models will identify the interstage CO_2 transfer rates and contactor-performance requirements necessary to achieve the predicted performance and compare these requirements against literature values and existing CNCE experimental observations. The model will also quantify the associated heating, cooling, condensation, separation, vacuum, and gas-drying requirements. The resulting CO_2 pressure, temperature, composition, and water content will define the inlet conditions for downstream compression or liquefaction. The assessment will therefore identify practical operating envelopes, quantify major energy flows, and establish performance requirements for downstream compression and storage systems.

Task 2.2 – CO_2 Storage, Cooling, and Energy-Storage Assessment

A screening-level assessment will determine whether gaseous or liquid CO_2 storage can provide sufficient load-shifting or cooling value to justify the additional capital and operating requirements relative to simpler alternatives. The analysis will estimate on-site CO_2 storage requirements, compression and liquefaction energy demands, gross cooling potential associated with liquid CO_2 expansion, internal process-cooling requirements, net cooling available for export, storage duration, round-trip energy efficiency, and integration requirements with existing data center cooling systems. The assessment will distinguish between once-through CO_2 destined for sequestration and CO_2 retained as the working inventory of a closed-loop cooling or energy-storage cycle. The objective is not to design a complete CO_2 energy-storage cycle, but rather to determine whether practical value exists for large-load cooling, energy storage, and load-shifting applications and identify the most promising operating concepts for future study.

Task 2.3 – Grid-Flexibility and Large-Load Integration Analysis

We will evaluate how effectively CO_2 recovery, compression, liquefaction, storage, and cooling operations can be scheduled to match diurnal and seasonal variations in waste-heat availability, renewable generation, electricity price, and cooling demand. Working with the energy-systems team, the assessment will examine:

- Load-shifting opportunities
- Peak-demand reduction potential
- Operational flexibility
- Cooling-demand management
- Interactions with renewable generation variability

Using representative large-load case studies, we will assess peak-demand reduction, operational flexibility, cooling-demand management, and integration with variable renewable generation.

Deliverables

D2.1: Process flow diagrams and mass/energy balance model, using initial bounding assumptions, describing staged waste-heat-assisted CO_2 concentration and recovery, compression, and liquid CO_2 production pathways (Month 2).

D2.2: Screening assessment of gaseous and liquid CO_2 storage pathways, including gross and net cooling potential, energy-storage opportunities, operating requirements, and performance estimates (Month 4).

D2.3: Grid-flexibility and large-load integration assessment identifying opportunities for load shifting, cooling-demand reduction, and system support (Month 5).

D2.4: Evaluation of low-cost compression and liquefaction pathways suitable for intermittent operation (Month 5).

Milestone

M2.1: Identify which individual and combined pathways for intermediate carbon storage, waste heat, CO_2 upgrading, and large-load integration may provide measurable load-management, cooling, or carbon-management benefits and establish priorities for future validation and development (Month 5).

Objective 3 – Preliminary Technoeconomic and Feasibility Assessment

Objective 3 will evaluate the economic viability of integrating ultra-low-energy atmospheric CO_2 capture, intermediate carbonate storage, waste-heat-assisted CO_2 upgrading, liquid CO_2 storage, large-load cooling, and load-management pathways. The objective is to identify the primary cost, energy, and operational drivers influencing system performance and determine whether the selected pathways warrant future development and validation.

Task 3.1 – Preliminary Technoeconomic Assessment

A preliminary technoeconomic assessment (TEA) will be conducted using process and performance data developed in Objectives 1 and 2. The analysis will estimate equipment costs, operating expenses, energy requirements, cooling benefits, carbon-management value, and load-management benefits associated with the selected pathways.

The assessment will focus on:

- Ultra-low-energy atmospheric CO_2 capture
- Intermediate carbonate storage
- Waste-heat-assisted CO_2 upgrading
- Compression and liquefaction systems

- Cooling and load-management pathways
- Operational flexibility benefits

Standard discounted-cash-flow methodologies will be used to estimate capital costs, operating costs, and economic performance under representative operating conditions.

Task 3.2 – Sensitivity and Uncertainty Analysis

A screening-level sensitivity and uncertainty analysis will be performed to identify the most important technical and economic variables affecting pathway feasibility. Parameters considered may include:

- Contactor productivity
- Sorbent working capacity
- Compression energy requirements
- Waste-heat availability
- Cooling-service value
- Electricity price
- Capital cost of process equipment
- Equipment utilization
- CO_2 recovery efficiency

The analysis will identify dominant cost and performance drivers, quantify uncertainty in projected performance, and establish performance targets required for future development.

Deliverables (D)

D3.1: Preliminary technoeconomic assessment estimating capital costs, operating costs, energy requirements, cooling benefits, and load-management value of the selected pathways (Month 4).

D3.2: Screening-level sensitivity and uncertainty analysis identifying dominant technical and economic drivers, plausible outcome ranges, and performance targets for future development (Month 5).

D3.3: Final feasibility report integrating process engineering, cooling, energy-storage, and technoeconomic findings and providing recommendations for future research and development priorities (Month 5).

Milestone (M)

M3.1: Identify the primary technical, economic, and operational barriers to deployment and establish quantitative performance targets required to justify future pilot-scale validation (Month 5).

High-Level Budget ($70,000)

- Avan Kumar, Process Engineering: $30,000
- Jose Manuel Dolot, Graduate Student, Energy Systems: $10,000

- Zeming Zhou, Graduate Student, Energy Systems: $10,000
- Leonardo Dayal, Graduate Student, CFD and Mechanical Engineering: $9,000
- Prototype materials and testing supplies: $5,000
- Computing/software resources: $2,000
- Travel and dissemination: $4,000

In-Kind Contributions

Meng Wu will provide in-kind faculty supervision and technical guidance related to energy systems, dispatch optimization, and large-load integration. Funds originally allocated for faculty effort are redirected to support a graduate student performing the detailed analysis.

Marcus Herrmann will provide in-kind oversight of CFD model development and validation activities.